



Move the Turtle — My Command Card

Name: _____

Python 3 • CrEAST 2026–27 • keep this on your desk!

Every turtle program starts the same way. Type these two lines first:

```
import turtle
t = turtle.Turtle()      # t is your pen / turtle
```

Move & Turn

Command	What it does
<code>t.forward(100)</code>	go forward 100 steps
<code>t.backward(100)</code>	go back 100 steps
<code>t.right(90)</code>	turn right 90°
<code>t.left(90)</code>	turn left 90°
<code>t.goto(x, y)</code>	jump to a point
<code>t.setheading(0)</code>	face a direction

Colour & Fill

Command	What it does
<code>t.color("red")</code>	pen colour
<code>t.fillcolor("yellow")</code>	colour inside a shape
<code>t.begin_fill()</code>	start colouring inside
<code>t.end_fill()</code>	stop colouring inside
<code>t.shape("turtle")</code>	turtle / arrow / circle

Pen up / down

Command	What it does
<code>t.penup()</code>	lift pen — move, no line
<code>t.pendown()</code>	drop pen — draw again
<code>t.pensize(5)</code>	make the line thick
<code>t.speed(5)</code>	0 = fastest, 10 = fast

Shapes & Finish

Command	What it does
<code>t.circle(50)</code>	draw a circle, radius 50
<code>t.dot(20)</code>	draw a filled dot
<code>t.write("Hi")</code>	write text
<code>t.hideturtle()</code>	hide the turtle
<code>turtle.done()</code>	last line — keep window open

Drawing a square — read it out loud

```
import turtle
t = turtle.Turtle()

for i in range(4):      # do it 4 times
    t.forward(100)      # one side
    t.left(90)          # one corner

turtle.done()
```

Notice 🧐

- `range(4)` = 4 sides → change to `range(3)` for a triangle (and turn 120).
- The two lines under `for` are **indented** (pushed in). That tells Python they belong to the loop.
- Try changing 100, the colour, or the turn angle.

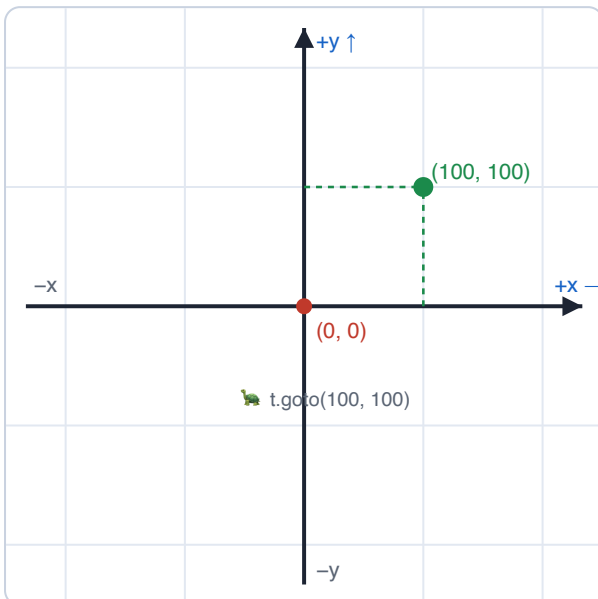
Experiment!



The Turtle Map & Loops

Where is the turtle? How do we repeat things?

The screen is a map — the middle is (0, 0)



- Middle of the screen is (0, 0).
- Go **right** → x gets **bigger** (+). Go **left** → x gets **smaller** (–).
- Go **up** → y gets **bigger** (+). Go **down** → y gets **smaller** (–).
- `t.goto(100, 100)` jumps to the green dot.
- Use `t.penup()` before `goto` if you don't want a line while jumping.

Try it: where is `t.goto(-100, -100)`? (Answer: bottom-left!)

Two ways to repeat — Loops

```
# for loop: when you know HOW MANY times
for i in range(5):
    print("Hello")

# prints Hello 5 times
```

```
# while loop: repeat UNTIL something changes
punch = 1
while punch <= 3:
    print("punch!")
    punch = punch + 1
```

Oops! 4 mistakes everyone makes

You wrote	Fix it to	Why
<code>print "Hi"</code>	<code>print("Hi")</code>	Python 3 needs the brackets ()
<code>for i in range(4)</code>	<code>for i in range(4):</code>	a loop / if needs a colon at the end
<code>for i in range(4):</code> <code>t.forward(100)</code>	<code>for i in range(4):</code> <code> t.forward(100)</code>	push the inside line in (indent)
<code>t.colour("red")</code>	<code>t.color("red")</code>	spell it the American way: color

Golden rule: the **last line** of a drawing program is always `turtle.done()` — without it, the window closes too fast to see your art!